$$C^{(21)}_{u\phi_1}{}^{i1\_i2\_} \to$$

$$\frac{1}{16\pi^2}\Big(\frac{1}{4} m_1 \tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{2,2,0}[m_1, \tilde\mu] + \frac{1}{12} m_2 \tilde\mu g_2^4 (17 y_u^{i1i2} - 4 y_u^{pi2}\delta_{pi1}) \mathrm{LF}_{2,2,0}[m_2, \tilde\mu] +$$

$$\frac{1}{6} m_2 \tilde\mu g_2^4 (-2 y_u^{i1i2} + y_u^{pi2}\delta_{pi1}) \mathrm{LF}_{3,2,-1}[m_2, \tilde\mu] -$$

$$\frac{1}{18}\tilde\mu g_1^2 y_d^{pr}\overline{a_d}^{pr}(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}) \mathrm{LF}_{2,2,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{36}\tilde\mu g_1^2 y_d^{pr}\overline{a_d}^{pr}(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$

$$\frac{1}{18}\tilde\mu g_1^2 y_e^{pr}\overline{a_e}^{pr}(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}) \mathrm{LF}_{2,2,0}[m_{\tilde e}^{r}, m_{\tilde l}^{p}] +$$

$$\frac{1}{36}\tilde\mu g_1^2 y_e^{pr}\overline{a_e}^{pr}(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde e}^{r}, m_{\tilde l}^{p}] +$$

$$\frac{1}{18}\tilde\mu y_e^{pr}\overline{a_e}^{pr}(-18 g_2^2 y_u^{i1i2} + g_1^2 (3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2})) \mathrm{LF}_{3,1,0}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] +$$

$$\frac{1}{18}\tilde\mu g_1^2 y_e^{pr}\overline{a_e}^{pr}(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] - \frac{1}{8}\tilde\mu y_e^{pr}\overline{a_e}^{pr}$$

$$(g_2^2(-19 y_u^{i1i2} + y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2})) \mathrm{LF}_{4,1,-1}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] +$$

$$\frac{1}{18}\tilde\mu y_e^{pr}\overline{a_e}^{pr}(3 g_2^2(-8 y_u^{i1i2} + y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{5,1,-2}[m_{\tilde l}^{p}, m_{\tilde e}^{r}] + \frac{1}{18}\tilde\mu y_d^{pr}\overline{a_d}^{pr}(-54 g_2^2 y_u^{i1i2} + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{3,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] + \frac{1}{18}\tilde\mu g_1^2 y_d^{pr}\overline{a_d}^{pr}(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}) \mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] +$$

$$\frac{1}{24}\tilde\mu y_d^{pr}\overline{a_d}^{pr}(9 g_2^2(19 y_u^{i1i2} - y_u^{si2}\delta_{si1}) - 5 g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] + \frac{1}{6}\tilde\mu y_d^{pr}\overline{a_d}^{pr}$$

$$(3 g_2^2(-8 y_u^{i1i2} + y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2})) \mathrm{LF}_{5,1,-2}[m_{\tilde q}^{p}, m_{\tilde d}^{r}] +$$

$$\frac{1}{36}\tilde\mu y_u^{pr}\overline{a_u}^{pr}(9 g_2^2(5 y_u^{i1i2} - y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{2,2,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] + \frac{3}{4}\tilde\mu g_2^2 y_u^{pr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$

$$\frac{1}{72}\tilde\mu y_u^{pr}\overline{a_u}^{pr}(9 g_2^2(5 y_u^{i1i2} - y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{3,2,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \frac{3}{4}\tilde\mu g_2^2 y_u^{pr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}] - \frac{1}{36}\tilde\mu y_u^{pr}\overline{a_u}^{pr}$$

$$(9 g_2^2(8 y_u^{i1i2} - y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2})) \mathrm{LF}_{3,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] -$$

$$\frac{1}{36}\tilde\mu y_u^{pr}\overline{a_u}^{pr}(9 g_2^2(5 y_u^{i1i2} - y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{3,2,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] - \frac{1}{12}\tilde\mu y_u^{pr}\overline{a_u}^{pr}$$

$$(9 g_2^2(-8 y_u^{i1i2} + y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2})) \mathrm{LF}_{4,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{6}\tilde\mu y_u^{pr}\overline{a_u}^{pr}(3 g_2^2(-8 y_u^{i1i2} + y_u^{si2}\delta_{si1}) + g_1^2(3 y_u^{i1i2} + y_u^{si2}\delta_{si1} - 4 y_u^{i1s}\delta_{si2}))$$

$$\mathrm{LF}_{5,1,-2}[m_{\tilde u}^{r}, m_{\tilde q}^{p}] - \frac{1}{2} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{3,1,0}[\tilde\mu, m_1] - \frac{7}{8} m_1\tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{3,2,-1}[\tilde\mu, m_1] -$$

$$\frac{1}{24} m_1\tilde\mu g_1^2 (3 g_2^2(-14 y_u^{i1i2} + y_u^{pi2}\delta_{pi1}) + g_1^2(3 y_u^{i1i2} + y_u^{pi2}\delta_{pi1} - 4 y_u^{i1p}\delta_{pi2}))$$

$$\mathrm{LF}_{4,1,-1}[\tilde\mu, m_1] + \frac{1}{2} m_1\tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde\mu, m_1] +$$

$$\frac{1}{18} m_1\tilde\mu g_1^2 (3 g_2^2(-8 y_u^{i1i2} + y_u^{pi2}\delta_{pi1}) + g_1^2(3 y_u^{i1i2} + y_u^{pi2}\delta_{pi1} - 4 y_u^{i1p}\delta_{pi2}))$$

$$\mathrm{LF}_{5,1,-2}[\tilde\mu, m_1] + \frac{1}{6} m_2\tilde\mu g_2^4 (-13 y_u^{i1i2} + 2 y_u^{pi2}\delta_{pi1}) \mathrm{LF}_{3,1,0}[\tilde\mu, m_2] +$$

$$\frac{1}{24} m_2\tilde\mu g_2^4 (-73 y_u^{i1i2} + 8 y_u^{pi2}\delta_{pi1}) \mathrm{LF}_{3,2,-1}[\tilde\mu, m_2] +$$

$$\frac{1}{8} m_2\tilde\mu g_2^2 (g_2^2(50 y_u^{i1i2} - 7 y_u^{pi2}\delta_{pi1}) - g_1^2(3 y_u^{i1i2} + y_u^{pi2}\delta_{pi1} - 4 y_u^{i1p}\delta_{pi2}))$$

$$\mathrm{LF}_{4,1,-1}[\tilde\mu, m_2] + 2 m_2\tilde\mu g_2^4 y_u^{i1i2} \mathrm{LF}_{4,2,-2}[\tilde\mu, m_2] +$$

$$\frac{1}{6} m_2\tilde\mu g_2^2 (3 g_2^2(-8 y_u^{i1i2} + y_u^{pi2}\delta_{pi1}) + g_1^2(3 y_u^{i1i2} + y_u^{pi2}\delta_{pi1} - 4 y_u^{i1p}\delta_{pi2}))$$

$$\mathrm{LF}_{5,1,-2}[\tilde\mu, m_2] - \frac{1}{36} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] -$$

$$\frac{1}{72} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] +$$

$$\frac{1}{36} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] - \frac{1}{6} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r}$$

$$y_u^{pi2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde d}^{r}, \tilde\mu] + \frac{1}{6} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde d}^{r}, \tilde\mu] +$$

$$\frac{1}{18} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] + \frac{1}{36} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r}$$

$$\mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] - \frac{1}{18} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] +$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde q}^{p}, \tilde\mu] - \frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r}$$

$$\mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde q}^{p}, \tilde\mu] + \frac{1}{72} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde d}^{r}] +$$

$$\frac{1}{36} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}] - \frac{1}{18} m_1\tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{1}{36} m_1\tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde q}^{i1}, \tilde\mu] - \frac{1}{36} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde u}^{i2}, m_{\tilde q}^{p}] +$$

$$\frac{1}{36} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde u}^{i2}, m_{\tilde q}^{i1}] - \frac{8}{9} m_1\tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{4}{9} m_1\tilde\mu g_1^4 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde u}^{i2}, \tilde\mu] + \frac{5}{24} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde d}^{r}] +$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde q}^{p}] +$$

$$\frac{1}{72} m_1\tilde\mu g_1^2 (3 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} - y_u^{i1i2}(11 g_1^2 + 3 g_2^2)) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde q}^{i1}] +$$

$$\frac{1}{18} m_1\tilde\mu g_1^2 (3 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} + y_u^{i1i2}(16 g_1^2 + 3 g_2^2)) \mathrm{LF}_{2,2,1,-1}[m_1, \tilde\mu, m_{\tilde u}^{i2}] -$$

$$\frac{3}{4} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{3}{4} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde q}^{p}, \tilde\mu] - \frac{3}{2} m_2\tilde\mu g_2^4 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_2, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{3}{4} m_2\tilde\mu g_2^4 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_2, m_{\tilde q}^{i1}, \tilde\mu] - \frac{3}{2}\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(m_1 + m_2) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_1] +$$

$$\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde\mu, m_1] +$$

$$\frac{1}{8} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde d}^{r}] +$$

$$\frac{1}{4} m_2\tilde\mu g_2^2 y_u^{pi2}(3 \overline{y_u}^{pr} y_u^{i1r} + g_2^2\delta_{pi1}) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde q}^{p}] +$$

$$\frac{1}{8} m_2\tilde\mu g_2^2 (3 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} + 16 g_2^2 y_u^{i1i2}) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde q}^{i1}] +$$

$$\frac{1}{4} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_2, \tilde\mu, m_{\tilde u}^{r}] +$$

$$\frac{2}{3} m_3\tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] +$$

$$\frac{1}{3} m_3\tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] -$$

$$\frac{2}{3} m_3\tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] +$$

$$\frac{2}{3} m_3\tilde\mu g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] +$$

$$\frac{1}{3} m_3\tilde\mu g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{2}{3} m_3\tilde\mu g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{1}{3} m_3\tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde q}^{i1}, m_{\tilde d}^{r}] +$$

$$\frac{1}{3} m_3\tilde\mu g_2^2 g_3^2 y_u^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}] -$$

$$\frac{1}{3} m_3\tilde\mu g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde u}^{i2}, m_{\tilde q}^{p}] +$$

$$\frac{1}{3} m_3\tilde\mu g_2^2 g_3^2 y_u^{i1i2} \mathrm{LF}_{2,2,1,-1}[m_3, m_{\tilde u}^{i2}, m_{\tilde q}^{i1}] - \frac{1}{4}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{4}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{8}\tilde\mu y_u^{pi2}\overline{a_d}^{pr}(g_2^2 y_d^{i1r} + y_d^{sr}\overline{y_u}^{st} y_u^{i1t}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde d}^{r}, \tilde\mu, m_{\tilde q}^{p}] -$$

$$\frac{3}{4}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, \tilde\mu] + \frac{1}{4}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, \tilde\mu] +$$

$$3\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde u}^{r}] -$$

$$3\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde u}^{r}] -$$

$$\frac{3}{4}\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pt} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{r}] -$$

$$\frac{3}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] +$$

$$\frac{3}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] -$$

$$3\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde d}^{t}] -$$

$$\frac{3}{8}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde q}^{s}] +$$

$$\frac{1}{8}\tilde\mu y_u^{pi2}\overline{a_d}^{pr}(g_2^2 y_d^{i1r} - y_d^{sr}\overline{y_u}^{st} y_u^{i1t}) \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, \tilde\mu, m_{\tilde d}^{r}] +$$

$$\frac{1}{8}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{p}, \tilde\mu, m_{\tilde u}^{r}] +$$

$$3\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pt} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$

$$3\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pt} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$

$$\frac{3}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{3}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{9}{8}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{18} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_1, m_{\tilde u}^{i2}] + \frac{1}{18} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{i1}, m_1, m_{\tilde u}^{i2}] +$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_1, \tilde\mu] + \frac{1}{4} m_2\tilde\mu g_2^4 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_2, \tilde\mu] +$$

$$\frac{2}{3} m_3\tilde\mu g_2^2 g_3^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^{i1}, m_3, m_{\tilde u}^{i2}] + \frac{2}{3} m_3\tilde\mu g_2^2 g_3^2 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^{i1}, m_3, m_{\tilde u}^{i2}] -$$

$$\frac{3}{2}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] +$$

$$\frac{3}{2}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] -$$

$$\frac{1}{8}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde u}^{r}, \tilde\mu, m_{\tilde q}^{p}] -$$

$$\frac{1}{18} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{i2}, m_1, m_{\tilde q}^{i1}] + \frac{1}{18} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{i2}, m_1, m_{\tilde q}^{i1}] -$$

$$\frac{2}{3} m_3\tilde\mu g_2^2 g_3^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,0}[m_{\tilde u}^{i2}, m_3, m_{\tilde q}^{i1}] + \frac{2}{3} m_3\tilde\mu g_2^2 g_3^2 y_u^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_{\tilde u}^{i2}, m_3, m_{\tilde q}^{i1}] +$$

$$\frac{1}{4}\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(m_1 - 5 m_2) \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_2] +$$

$$\frac{3}{2} m_2\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{4,1,1,-2}[\tilde\mu, m_1, m_2] - \frac{1}{2} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde d}^{r}] +$$

$$\frac{1}{4} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde d}^{r}] -$$

$$\frac{1}{2} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde q}^{p}] +$$

$$\frac{1}{4} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{p}] - \frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2}$$

$$\mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde q}^{i1}] + \frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{i1}] -$$

$$\frac{1}{3} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_{\tilde u}^{i2}] + \frac{1}{3} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r}$$

$$\mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_{\tilde u}^{i2}] + \frac{1}{2}\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(m_1 + 3 m_2) \mathrm{LF}_{3,2,1,-2}[\tilde\mu, m_2, m_1] -$$

$$\frac{1}{2} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde d}^{r}] + \frac{1}{4} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2}$$

$$\mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde d}^{r}] - \frac{3}{2} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde q}^{p}] +$$

$$\frac{3}{4} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{p}] - \frac{3}{4} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2}$$

$$\mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde q}^{i1}] + \frac{3}{4} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{i1}] -$$

$$m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_2, m_{\tilde u}^{r}] + \frac{1}{2} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_2, m_{\tilde u}^{r}] +$$

$$\frac{1}{4}\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$

$$\frac{1}{4}\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t}\overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{4}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$

$$\frac{1}{4}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$

$$\frac{1}{6}\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(m_1 + m_2) \mathrm{LF}_{1,1,1,1,0}[m_1, m_2, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{1}{12} m_2\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_2, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{1}{8}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{1}{16}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{36}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$

$$\frac{1}{72}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$

$$\frac{2}{9} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}] -$$

$$\frac{1}{8} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{1}{48} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, \tilde\mu] -$$

$$\frac{2}{3} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{1}{2}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{1}{4}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{4}{9}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}] -$$

$$\frac{2}{9}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}] -$$

$$\frac{1}{2} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, \tilde\mu] -$$

$$\frac{2}{9} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{1}{6} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{2}{9}\tilde\mu g_1^4 m_1^2 a_u^{i1i2} \mathrm{LF}_{2,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_1, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{3}{8}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{3}{16}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{4}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$

$$\frac{1}{8}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$

$$\frac{3}{8} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{1}{16} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{3}{4}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{3}{8}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{1}{2}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$

$$\frac{1}{4}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$

$$\frac{1}{2} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{3}{4} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{1}{8} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{4}{3}\tilde\mu g_3^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$

$$\frac{2}{3}\tilde\mu g_3^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}] -$$

$$\frac{8}{3} m_3\tilde\mu g_3^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}] +$$

$$\frac{4}{3}\tilde\mu g_3^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}] -$$

$$\frac{2}{3}\tilde\mu g_3^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}] -$$

$$\frac{8}{3} m_3\tilde\mu g_3^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{1,1,1,1,0}[m_3, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{1}{48}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_1, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{1}{48} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_1, m_{\tilde q}^{i1}, \tilde\mu] -$$

$$\frac{1}{16}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_2, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{1}{16} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_2, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{1}{2}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1r} y_u^{si2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{1}{4}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1r} y_u^{si2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{2}\tilde\mu \overline{y_u}^{st}\overline{a_d}^{pr}(2 y_d^{i1r} y_u^{pt} y_u^{si2} + y_d^{sr} y_u^{pi2} y_u^{i1t}) \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, \tilde\mu] -$$

$$\frac{3}{4}\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{r}, m_{\tilde q}^{s}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] +$$

$$\frac{1}{4}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1r} y_u^{si2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde d}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\tilde\mu \overline{y_d}^{st} y_d^{i1t} y_u^{pi2} y_u^{sr}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{3}{4}\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] -$$

$$\frac{1}{4}\tilde\mu^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2}\overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{r}, m_{\tilde l}^{s}, m_{\tilde e}^{t}, m_{\tilde l}^{p}] +$$

$$\frac{1}{4}\tilde\mu^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2}\overline{a_e}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^{t}, m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde l}^{p}] -$$

$$\frac{1}{2}\tilde\mu^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2}\overline{a_e}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{p}] +$$

$$\frac{1}{2}\tilde\mu^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} y_u^{i1i2}\overline{a_e}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde l}^{s}, m_{\tilde e}^{r}, m_{\tilde e}^{t}, m_{\tilde l}^{p}] +$$

$$\frac{1}{72}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] +$$

$$\frac{1}{48}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde d}^{r}, \tilde\mu] -$$

$$\frac{1}{12}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_1, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{1}{8}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] +$$

$$\frac{1}{16}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde d}^{r}, \tilde\mu] +$$

$$\frac{1}{4}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] +$$

$$\frac{1}{8}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_2, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{2}{3}\tilde\mu g_3^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] -$$

$$\frac{1}{4}\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{r}, m_{\tilde q}^{s}, \tilde\mu] -$$

$$3\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] +$$

$$3\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde d}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] -$$

$$\frac{3}{4}\tilde\mu y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde u}^{t}] +$$

$$\frac{9}{4}\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde d}^{t}, m_{\tilde u}^{r}] -$$

$$\frac{3}{4}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] +$$

$$\frac{3}{4}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] +$$

$$\frac{1}{2}\tilde\mu \overline{y_u}^{st} y_u^{pi2} y_u^{sr} y_u^{i1t}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{1}{4}\tilde\mu \overline{y_u}^{st} y_u^{pi2} y_u^{sr} y_u^{i1t}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$3\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{s}] -$$

$$\frac{3}{8}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] +$$

$$\frac{1}{2}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}, \tilde\mu] +$$

$$\frac{3}{8}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{p}, m_{\tilde u}^{t}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] -$$

$$\frac{3}{2}\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] +$$

$$\frac{3}{2}\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] +$$

$$3\tilde\mu y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] -$$

$$3\tilde\mu y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] +$$

$$\frac{1}{4}\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde d}^{r}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{3}{4}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] +$$

$$\frac{3}{4}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] +$$

$$\frac{1}{4}\tilde\mu \overline{y_u}^{st} y_u^{pi2} y_u^{sr} y_u^{i1t}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{9}{8}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}] -$$

$$3\tilde\mu y_d^{sr} y_u^{i1i2}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{3}{8}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^{s}, m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}] -$$

$$\frac{1}{72}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{48} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_1, m_{\tilde d}^{r}, \tilde\mu] -$$

$$\frac{1}{8}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{16} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde d}^{r}, \tilde\mu] -$$

$$\frac{1}{4}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{1}{8} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_2, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{2}{3}\tilde\mu g_3^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde q}^{i1}, m_3, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{2}{9}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] +$$

$$\frac{1}{12}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_1, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{1}{8}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_2, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{1}{8} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_2, m_{\tilde q}^{i1}, \tilde\mu] +$$

$$\frac{2}{3}\tilde\mu g_3^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_3, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{3}{2}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] +$$

$$\frac{3}{2}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}] -$$

$$\frac{1}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{t}, \tilde\mu] +$$

$$\frac{3}{4}\tilde\mu y_u^{sr} y_u^{i1i2}\overline{a_u}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde u}^{r}, m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] +$$

$$\frac{1}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{t}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{2}{9}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{i2}, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{1}{12} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{i2}, m_1, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{2}{3}\tilde\mu g_3^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[m_{\tilde u}^{i2}, m_3, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{1}{24}\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(13 m_1 - 3 m_2) \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_2, m_{\tilde q}^{i1}] +$$

$$\frac{2}{3} m_2\tilde\mu g_1^2 g_2^2 y_u^{i1i2} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_2, m_{\tilde u}^{i2}] +$$

$$\frac{23}{48}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{2} m_1 g_1^2\tilde\mu^3 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,0}[\tilde\mu, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{16} m_1\tilde\mu g_1^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] -$$

$$\frac{7}{12}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{1}{4} m_1\tilde\mu g_1^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{9}{16}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{1}{2} m_2 g_2^2\tilde\mu^3 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,0}[\tilde\mu, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{p}] +$$

$$\frac{3}{16} m_2\tilde\mu g_2^2 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}] -$$

$$\frac{5}{8}\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{p}, m_{\tilde u}^{r}] +$$

$$\frac{3}{8} m_2\tilde\mu g_2^2 \overline{y_u}^{pr} y_u^{pi2} y_u^{i1r} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{r}] -$$

$$\frac{1}{4}\tilde\mu \overline{y_d}^{st} y_d^{pt} y_d^{i1r} y_u^{si2}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}] -$$

$$\frac{1}{4}\tilde\mu y_d^{sr}\overline{y_u}^{st} y_u^{pi2} y_u^{i1t}\overline{a_d}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}] -$$

$$\frac{1}{4}\tilde\mu \overline{y_u}^{st} y_u^{pi2} y_u^{sr} y_u^{i1t}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}] -$$

$$\frac{1}{4}\tilde\mu \overline{y_u}^{st} y_u^{pt} y_u^{si2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{2,1,1,1,-1}[\tilde\mu, m_{\tilde q}^{p}, m_{\tilde u}^{r}, m_{\tilde u}^{t}] +$$

$$\frac{2}{3}\tilde\mu g_1^2 g_2^2 y_u^{i1i2}(m_1 + m_2) \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{2}{3} m_1 m_2\tilde\mu g_1^2 g_2^2 a_u^{i1i2} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_2, m_{\tilde q}^{i1}, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{2}{3} m_1\tilde\mu g_1^2 y_d^{i1r}(-\overline{a_d}^{pr} a_u^{pi2} + \tilde\mu^2 \overline{y_d}^{pr} y_u^{pi2}) \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, \tilde\mu] +$$

$$\frac{2}{9} m_1\tilde\mu g_1^2 y_d^{i1r}(-\overline{a_d}^{pr} a_u^{pi2} + \tilde\mu^2 \overline{y_d}^{pr} y_u^{pi2}) \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] +$$

$$\frac{1}{6}\tilde\mu g_1^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{6} m_1 g_1^2\tilde\mu^3 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\frac{2}{3}\tilde\mu g_1^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_1, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{1}{6} m_1\tilde\mu g_1^2 y_u^{pi2}\overline{a_u}^{pr} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{2}{9} m_1\tilde\mu g_1^2 y_u^{pi2}\overline{a_u}^{pr} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_1, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}] -$$

$$\frac{1}{2}\tilde\mu g_2^2 y_d^{i1r} y_u^{pi2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] +$$

$$\frac{1}{2} m_2 g_2^2\tilde\mu^3 \overline{y_d}^{pr} y_d^{i1r} y_u^{pi2} \mathrm{LF}_{1,1,1,1,1,0}[m_2, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, \tilde\mu] -$$

$$\tilde\mu g_2^2 y_u^{pi2} y_u^{i1r}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,-1}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] -$$

$$\frac{1}{2} m_2\tilde\mu g_2^2 y_u^{pi2}\overline{a_u}^{pr} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_2, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{r}, \tilde\mu] +$$

$$\frac{8}{3} m_3\tilde\mu g_3^2 y_d^{i1r}(-\overline{a_d}^{pr} a_u^{pi2} + \tilde\mu^2 \overline{y_d}^{pr} y_u^{pi2}) \mathrm{LF}_{1,1,1,1,1,0}[m_3, m_{\tilde d}^{r}, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}] -$$

$$\frac{8}{3} m_3\tilde\mu g_3^2 y_u^{pi2}\overline{a_u}^{pr} a_u^{i1r} \mathrm{LF}_{1,1,1,1,1,0}[m_3, m_{\tilde q}^{i1}, m_{\tilde q}^{p}, m_{\tilde u}^{i2}, m_{\tilde u}^{r}] -$$

$$\tilde\mu^3 \overline{y_d}^{st} y_d^{pt} y_d^{i1r} y_u^{si2}\overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, \tilde\mu] +$$

$$\tilde\mu y_d^{i1r} y_u^{si2}\overline{a_d}^{pr}\overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{r}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{t}, \tilde\mu] -$$

$$\tilde\mu^3 \overline{y_d}^{st} y_d^{i1t} y_u^{pi2} y_u^{sr}\overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde d}^{t}, m_{\tilde q}^{p}, m_{\tilde q}^{s}, m_{\tilde u}^{r}, \tilde\mu]\Big)$$